

$$\begin{aligned}
\mu_{i,t} &= \beta_0 + f(z_i) + \beta_1 \text{GL12}_{i,t} \\
&\quad + [\beta_2 + b_2(\mathbf{s}_i)] \sin\left(\frac{2\pi \text{DOY}_{i,t}}{365}\right) \\
&\quad + [\beta_3 + b_3(\mathbf{s}_i)] \cos\left(\frac{2\pi \text{DOY}_{i,t}}{365}\right) \\
&\quad + \omega(\mathbf{s}_i) + \epsilon_t(\mathbf{s}_i), \\
\epsilon_t(\mathbf{s}) &= \rho \epsilon_{t-1}(\mathbf{s}) + \sqrt{1 - \rho^2} \eta_t(\mathbf{s}) \\
\eta_t(\mathbf{s}) &\sim \mathcal{GP}(0, \Sigma) \\
y_{i,t} &\sim t_\nu(\mu_{i,t}, \sigma)
\end{aligned} \tag{1}$$

Definitions

- $y_{i,t}$ is the response variable, which is the difference between observations and **GL12** temperature at location i and year t .
- β_0 is the intercept.
- β_1 is the fixed effect of **GL12** temperature.
- z_i is the depth (in log space) at location i .
- β_2 and β_3 are the population-level seasonal effects.
- $b_2(\mathbf{s}_i)$ and $b_3(\mathbf{s}_i)$ are spatially varying coefficients for the seasonal terms on day of year $\text{DOY}_{i,t}$.
- $\omega(\mathbf{s}_i)$ is the spatial Gaussian random field.
- $\epsilon_t(\mathbf{s})$ is the spatiotemporal random field at location $\mathbf{s}$ and time t .
- ρ is the AR(1) temporal correlation parameter.
- $\eta_t(\mathbf{s})$ is the innovation spatial field, representing new spatial variation not explained by the spatiotemporal field in the previous time step.
- $\mathcal{GP}(0, \Sigma)$ denotes a Gaussian process with mean zero and covariance matrix Σ .
- Σ is the Matérn covariance structure represented through the SPDE approximation.
- σ is the scale parameter of the Student- t distribution.
- ν is the degrees-of-freedom parameter controlling tail heaviness.

Methods We modeled D using a spatial Gaussian mixed-effects model with an identity link. Fixed effects included Glor , $\sin(\text{DOY})$, and $\cos(\text{DOY})$. Spatially varying coefficients were estimated for the seasonal terms $\sin(\text{DOY})$ and $\cos(\text{DOY})$, allowing the magnitude and phase of seasonal patterns to vary geographically. A spatial Gaussian random field was included to account for residual spatial dependence, and a spatiotemporal random field with an $\text{AR}(1)$ temporal structure was fitted across years using the SPDE approach. The Student- t distribution was used to accommodate heavy-tailed residual variation and reduce sensitivity to outlying observations.